

BE I Year 'CSE-A & CSE-B' Class Test-III
1MERC3: Elements of Mechanical Engineering, 2023

Duration: 70 Min.

Max. Marks: 20

Note: Attempt all questions.

Q1 An engine working on otto cycle has a volume of 0.45 m^3 , pressure 1 bar and temperature 30°C at beginning of compression stroke. At the end of compression stroke, the pressure is 11 bar. And 210 KJ of heat is added at constant volume. Determine (i) pressures, temperatures and volumes at salient points in cycle. (ii) Percentage clearance (iii) efficiency. 06

Q2 Derive the efficiency of an air standard diesel cycle. And calculate efficiency if cycle compression ratio is 15 and cut off is 6% of stroke. 06

Q3 Explain Moulding process and Explain any 5 casting defects (2+2)

Q4 Explain Importance of getting & risering (4)